THE MASTERMIND COACHING CLASSES

Revision Notes

Class 8 · Science

NCERT Curiosity — Chapters 8 to 13 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

PART 2 — CHAPTERS 8 to 13. Read each chapter's numbered sections in the textbook's order, learn the bold keywords and tables, revise the definition boxes, practise the activities, then test yourself with the Quick Questions and the Key Takeaways.

CH.	CHAPTER TITLE	CORE THEME
8	Nature of Matter: Elements, Compounds and Mixtures	Mixtures, pure substances, minerals
9	The Amazing World of Solutes, Solvents and Solutions	Solutions, solubility, density
10	Light: Mirrors and Lenses	Spherical mirrors, laws of reflection, lenses
11	Keeping Time with the Skies	Moon phases, calendars, satellites
12	How Nature Works in Harmony	Ecosystems, food chains, balance
13	Our Home Earth, a Unique Life-Sustaining Planet	Why Earth supports life, reproduction, threats

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CHAPTER 08

Nature of Matter: Elements, Compounds and Mixtures

*Everything around us — the staircase, the air, the water in a bottle, the food in a lunch box — is **matter**, and most of it is not a single substance but two or more substances together. This chapter shows how scientists sort matter into **mixtures** and **pure substances**, and how pure substances divide again into **elements** and **compounds**. Two experiments (splitting water with electricity and heating iron with sulfur) carry most of the ideas.*

Big Questions

- Which things in the world are made of matter and which are not?
- How can elements be combined to make a compound?
- How could a compound that absorbs carbon dioxide from the air help with environmental problems?

8.1 What Are Mixtures?

Poha, a sprout salad, soup, lemonade and sugar dissolved in water are all mixtures: two or more substances are mixed and **each keeps its own properties**.

DEFINITION

A **mixture** is made when two or more substances are mixed so that each substance **retains its properties**. The substances in it are its **components**. The components **do not react chemically** with each other.

Kind of mixture	What you see	Examples
Non-uniform (heterogeneous)	Components can be seen separately by the naked eye or with a magnifier	Sprout salad (green gram, chickpea, onion, tomato), poha, sand and water, muddy water
Uniform (homogeneous)	Components are evenly spread and cannot be told apart, even under a microscope	Sugar in water, air, vinegar, seawater, soda water, stainless steel

▸ Alloys — uniform mixtures of metals

- **Stainless steel** = iron + nickel + chromium + a little carbon, so evenly blended that it looks the same throughout.
- **Brass** = copper + zinc; **bronze** = copper + tin.
- Alloys are usually stronger and more durable than the pure metal (stainless steel versus pure iron).

REMEMBER

Our scientific heritage: ancient India called metal blends *mishraloha*. Texts such as the Charaka Samhita, Susruta Samhita and Rasaratna Samucchaya mention alloys used as medicine. Bronze (*Kamsya*) was made of 4 parts copper (*Tamra*) and 1 part tin (*Vanga*) and was believed to aid digestion and immunity.

▸ 8.1.1 Is air a mixture?

Yes — air is a **uniform mixture** of mainly nitrogen (about **78%**), oxygen, argon, carbon dioxide and water vapour. Oxygen supports life and combustion; nitrogen does **not** take part in combustion. Warm air touching a cool surface loses its water vapour as tiny droplets, which shows water vapour is present.

Activity 8.1 — Showing carbon dioxide in air (lime water)
Add a little **calcium oxide (quick lime)** slowly to half a tumbler of water — it reacts vigorously and gives out heat (take care). Stir to make **calcium hydroxide solution = lime water**; filter it so it is colourless. Leave it in a petri dish for a few hours, stirring now and then. **Observation:** it turns **milky**.

Why it turns milky
Carbon dioxide from the air reacts with calcium hydroxide to give insoluble tiny white particles of **calcium carbonate** and water: *calcium hydroxide + carbon dioxide → calcium carbonate + water*. So the experiment proves CO_2 is present in air.

Activity 8.2 — Dust in air
Leave a clean black sheet of paper near an open window for a few hours and look with a magnifier: tiny **dust particles** settle on it. Dust is **suspended** in air, is **not a real part of air**, varies from place to place and time to time, and counts as a **pollutant**.

> **NOTE**
> Major air pollutants: **particulate matter** (dust, soot) and gases — carbon monoxide, ozone, nitrogen dioxide, sulfur dioxide. The **Air Quality Index (AQI)** describes air quality.

▸ 8.1.2 Types of mixtures (by physical state of the components)

Type	Examples	Uniform / non-uniform
Gas + gas	Air	Uniform
Gas + liquid	Aerated (soda) water; oxygen dissolved in water	Uniform
Solid + gas	Carbon (soot) particles in air	Non-uniform
Liquid + liquid	Acetic acid in water (vinegar) — uniform; oil and water — non-uniform	Both kinds
Solid + liquid	Sand and water — non-uniform; seawater — uniform	Both kinds
Solid + solid	Baking powder (baking soda + tartaric acid); alloys	Alloys are uniform; baking powder is a solid mixture

- In everyday life we separate a mixture only to **get the part we want** and throw the rest away.
- In science the aim of separation is to **obtain pure substances**.
- In science every component of a mixture must itself be a **pure substance**, though in common speech a mixture's parts may be mixtures (poha, salad).

8.2 What Are Pure Substances?

The word *pure* means different things in daily life and in science.

	Common usage	Science
Meaning of ‘pure’	Not adulterated — nothing cheaper or poorer added	Contains **only one kind of substance / one kind of particle**
Example	‘Pure’ milk, ghee, spices	Distilled water, sugar, gold, oxygen
Is milk pure?	Yes, if unadulterated	No — it is a mixture of many substances

- **Adulteration** is the illegal addition of cheaper or poor-quality substances to increase quantity or cut cost; it lowers quality and can harm health.
- A **pure substance** is a kind of matter that **cannot be separated into other kinds of matter by any physical process**; all its particles are of the same type and behave identically.
- Milk, packed fruit juice and soil are **mixtures**; baking soda and sugar are **pure substances**.

8.3 Types of Pure Substances

Cooling water gives ice and boiling gives vapour, and we can get water back — the particles stay the same. Passing electricity is different: it **changes** the water.

Activity 8.3 — Passing electricity through water (teacher demonstration)
Fill a beaker 2/3 with water and add a few drops of **dilute sulfuric acid**. Fill two small test tubes with this water and invert each over a terminal of a **9 V battery** placed in the beaker (do not use a lithium-ion battery). After 10–15 minutes gas has collected in both tubes — **unequal volumes**, more in one tube than the other (about 2 : 1 hydrogen to oxygen).

Testing the gases
Bring a burning candle near each tube. One gives a **pop** sound = **hydrogen**. In the other the flame **glows brighter** = **oxygen**. The gases are not water vapour, since vapour would have condensed back.

Inference
Water is made of two different constituents: *water → hydrogen + oxygen* (a chemical change).

▸ 8.3.1 Elements

- Hydrogen and oxygen are **elements** — pure substances made of identical particles called **atoms**, different from the atoms of every other element.
- An **element cannot be broken down into simpler substances**; elements are the **building blocks** of all matter (gold, silver, sulfur, carbon…).
- Atoms of most elements cannot exist alone; two or more atoms join to form a stable **molecule** — e.g. two hydrogen atoms make one hydrogen molecule, two oxygen atoms make one oxygen molecule.
- Elements are classified as **metals** (gold, silver, magnesium, iron, aluminium) and **non-metals** (carbon, sulfur, hydrogen, oxygen). **Metalloids** such as silicon and boron have properties in between.

IMPORTANT

Facts on elements: 118 elements are known and most are solids. **Eleven** are gases at room temperature (all non-metals, e.g. oxygen, helium, nitrogen). Only **two are liquids** at room temperature: **mercury** (metal) and **bromine** (non-metal). Gallium and caesium are solids that turn liquid at about **30 °C (303 K)**. A mobile phone uses more than **45 different elements** (aluminium, copper, silicon, cobalt, lithium, gold, silver…).

▸ 8.3.2 Compounds

- In water, hydrogen and oxygen are so tightly attached that **no physical method** can separate them — that is why water is a **compound**.
- **Compounds** are formed when different elements combine **chemically in a fixed ratio** to give something **entirely new**; the compound's properties are **different from those of its elements**; its elements **cannot be separated by physical methods**.
- **Water:** H : O atoms in the ratio **2 : 1**. Hydrogen is a fuel, oxygen supports burning, but water puts out fire.
- **Sodium chloride (common salt):** soft metal sodium + hazardous gas chlorine combine in a **1 : 1** ratio to give a harmless, tasty, life-essential substance. Evaporation separates salt from *salt solution*, but no physical process splits salt into sodium and chlorine.

Activity 8.4 — Heating sugar (with teacher)
Heat a teaspoon of sugar gently in a boiling tube. It turns **brown**, then **black (charred)**; **water droplets** appear near the open end; **charcoal (carbon)** is left, which burns like coal.

Inference
The water must have come from the sugar (not from the air, as the tube is being heated). Sugar decomposes into carbon and water, so it is **not an element but a compound of carbon, hydrogen and oxygen**.

▸ Activity 8.5 — Mixture versus compound (iron filings + sulfur)

Take **5.6 g iron filings** and **3.2 g sulfur powder**, mix well = **Sample A**. Heat half of it in a china dish with stirring until a **black mass** forms, cool and grind = **Sample B**.

Test	Sample A (mixture)	Sample B (compound, iron sulfide)
Appearance	Black iron and yellow sulfur seen separately — non-uniform	Same colour and texture throughout (black)
Magnet	**Iron is attracted**, so iron and sulfur can be separated	**Not attracted**; the elements can no longer be separated
Dilute HCl	Iron reacts: iron chloride + **hydrogen** (colourless, odourless, burns with a *pop*); sulfur stays as yellow solid	Iron chloride + **hydrogen sulfide** (colourless, **rotten-egg smell**)
Properties of parts	Each component keeps its own properties	Completely new properties — a new substance formed
Type of change	Physical mixing	Chemical change by heating

- Equations in words: *iron + dilute hydrochloric acid → iron chloride + hydrogen*; *iron sulfide + dilute hydrochloric acid → iron chloride + hydrogen sulfide*.
- Safety: demonstrate in a fume hood or ventilated area, never smell a gas directly — **waft** the fumes towards the nose.

Feature	Element	Compound	Mixture
Made of	One kind of atom	Two or more elements chemically combined	Two or more substances physically mixed
Composition	Fixed	**Fixed ratio**	Variable

Feature	Element	Compound	Mixture
Properties	Its own	**Different from its elements**	Components keep their properties
Separation	Cannot be broken down further	Only by chemical methods	By physical methods
Pure?	Yes	Yes	No
Examples	Gold, oxygen, iron, sulfur	Water, sugar, salt, iron sulfide	Air, brass, seawater, soil

8.4 How Do We Use Elements, Compounds and Mixtures?

- Air is a **mixture**; water is a **compound**; iron and aluminium (**elements**) build bridges, buildings and vehicles.
- Chemists study how elements combine into compounds to make **medicines, vaccines and fertilisers** (better crop yields).
- Engineers design materials with special properties — **alloys** like stainless steel; wood, steel and concrete are mixtures.
- **Graphene aerogel** (carbon) is among the lightest materials on earth, highly porous and absorbent, so it may help **clean up oil spills**, make energy-saving devices and coat buildings.

8.5 What Are Minerals?

- Most **rocks are mixtures of minerals**, which can be seen with the eye, a magnifier or a microscope.
- **Minerals** are natural, solid substances found in the Earth with a **fixed chemical composition**. Most are **compounds**; a few are pure elements called **native minerals** (gold, silver, copper, sulfur, carbon).
- Examples: **quartz, calcite, mica, pyroxene, olivine, talc**.
- Uses: **cement** is made from calcite, quartz, alumina and iron oxide; **talcum powder** from talc.
- Gold is both a **metal** (by type) and a **mineral** (by occurrence in nature).

> **REMEMBER**
>
> **Dhokra art** (Bihar and Odisha): a design is shaped in beeswax, covered with clay to make a mould, the wax is melted out and **molten brass or bronze** is poured in. It shows a practical use of alloys in Indian craft.

What Is and Is Not Matter

Matter is anything that has **mass and occupies space**; elements and compounds are its building blocks. But **light, heat, electricity, thoughts and emotions are not matter**, even though they are important parts of our world.

Quick Questions with Model Answers

Likely question	Model answer
Define a mixture and name its two types.	Two or more substances mixed with each keeping its own properties and no chemical reaction; uniform and non-uniform.

Likely question	Model answer
How does the scientific meaning of 'pure' differ from everyday meaning?	Everyday: unadulterated. Science: contains only one kind of substance whose particles are all identical.
How do we show that air contains carbon dioxide?	Lime water left open to air turns milky because carbon dioxide forms insoluble calcium carbonate.
What happens when electricity passes through acidified water? How are the gases identified?	Water breaks into hydrogen and oxygen. Hydrogen burns with a pop; oxygen makes a candle flame brighter.
Why is water called a compound?	Hydrogen and oxygen are chemically combined in a fixed 2 : 1 ratio and cannot be separated by physical methods; water's properties differ from both.
Why is sugar not an element?	On heating it breaks into carbon and water, so it contains carbon, hydrogen and oxygen.
Distinguish iron-sulfur mixture and iron sulfide.	The mixture keeps the properties of both, is magnetic and separable; iron sulfide is a new black substance, not attracted by a magnet and gives hydrogen sulfide with HCl.
Name the two elements liquid at room temperature.	Mercury (metal) and bromine (non-metal).
What are alloys? Give two examples with their components.	Uniform mixtures of metals; brass (copper + zinc), bronze (copper + tin), stainless steel (iron, nickel, chromium, carbon).
What are native minerals?	Minerals that are pure elements, such as gold, silver, copper, sulfur, carbon.
Can a substance be both an element and a compound?	No. An element has only one kind of atom; a compound has two or more elements combined.

★ KEY TAKEAWAYS

- A **mixture** keeps the properties of its components; uniform mixtures look the same throughout, non-uniform ones show their parts.
- A **pure substance** has identical particles and cannot be separated physically — it is an **element** or a **compound**.
- **Elements** are the simplest substances (118 known); **compounds** are elements combined chemically in a **fixed ratio** with new properties.
- Water = H : O in **2 : 1**; salt = Na : Cl in **1 : 1**; sugar = carbon, hydrogen, oxygen.
- Iron + sulfur **mixed** gives a separable mixture; **heated** gives iron sulfide, a compound.
- **Minerals** are natural solids of fixed composition — mostly compounds, a few native elements; rocks are mixtures of minerals.
- Light, heat, electricity, thoughts and emotions are **not matter**.

CHAPTER 09

The Amazing World of Solutes, Solvents and Solutions

*Every sip of homemade ORS tastes the same because sugar and salt spread evenly through the water. This chapter explains **solutions** (uniform mixtures), how much solute a solvent can hold (**solubility**, saturated and unsaturated solutions), how temperature changes it, and then **density** — the property that decides, with other factors, whether things float or sink — including how to measure mass, volume and density.*

Big Questions

- What happens when you add too much sugar to tea and it stops dissolving? How can you solve this?
- Why do sugar and salt dissolve in water but not in oil? Why is water a good solvent?
- Why are water bottles usually tall and cylindrical instead of spherical?

9.1 Solute, Solvent and Solution

Mixture	Type	Why
Sugar + salt + water (ORS)	Uniform	Components evenly distributed, every sip same
Chalk powder / sand / sawdust + water	Non-uniform	Components not evenly spread; can be seen

DEFINITION

A **solution** is a uniform mixture. When a solid dissolves in a liquid, the solid is the **solute** and the liquid is the **solvent**: *solute* + *solvent* → *solution*. In a solution of **two liquids**, the one in **smaller amount is the solute** and the one in larger amount is the solvent.

- Components of a solution cannot be seen separately; in non-uniform mixtures they can be seen with the eye or a magnifier.
- A mixture of gases (such as air) is also a uniform mixture, i.e. a solution.
- **Gulab jamun chashni:** a large amount of sugar in a small amount of water — still water is the solvent and sugar the solute.

9.2 How Much Solute Can a Fixed Amount of Solvent Dissolve?

Activity 9.1 — Add salt to water
Half-fill a tumbler with water; add one spoon of salt, stir till dissolved, keep adding spoon by spoon and stir. Record in a table whether each spoon dissolves.

Observation
At first salt dissolves completely; after a few spoons some **settles at the bottom**. Water has reached its **limit** at that temperature.

Term	Meaning
Unsaturated solution	More solute can still be dissolved at that temperature
Saturated solution	No more solute dissolves at that temperature; extra settles at the bottom

Term	Meaning
Concentration	Amount of solute present in a fixed quantity of solution (or solvent)
Dilute solution	Relatively less solute
Concentrated solution	Relatively more solute (dilute and concentrated are *relative* terms)
Solubility	Maximum amount of solute that dissolves in a fixed quantity (100 mL / 100 g) of solvent at a given temperature

REMEMBER

4 spoons of salt in 50 mL of water is **more concentrated** than 2 spoons in 100 mL — compare the solute per same volume.

▸ 9.2.1 Effect of temperature on solubility

Activity 9.2 — Baking soda in water (demonstration)
Take about 50 mL of water at ~20 °C and add baking soda until some is left undissolved. Heat to 50 °C, stirring: the undissolved soda dissolves. Add more until some remains, heat to 70 °C: it dissolves again.

Inference
Water at 70 °C dissolves more than at 50 °C, which dissolves more than at 20 °C. For **most solids, solubility increases with temperature**; a saturated solution becomes **unsaturated** when heated.

NOTE

Scientific heritage: water is the main solvent in Ayurveda and Siddha formulations; hydro-alcoholic extracts, oils, ghee and milk are also used as solvents. **Asima Chatterjee** developed anti-epileptic and anti-malarial drugs by extracting compounds from medicinal plants with solvents; she was the second Indian woman to earn a Doctorate of Science, first woman to get the Shanti Swarup Bhatnagar Award in chemical science, and received the Padma Bhushan.

9.3 Solubility of Gases

- Many gases, including oxygen, dissolve in water (a uniform mixture), but only to a small extent. This **dissolved oxygen sustains aquatic life** — plants, fishes and others.
- **For gases, solubility generally decreases as temperature increases.** Cold water holds more oxygen; when water warms, less oxygen stays dissolved, which can harm aquatic life.

Solute	Effect of rising temperature on solubility in liquid
Solids (most)	**Increases**
Gases	**Decreases**

9.4 Why Do Objects Float or Sink in Water?

Husk floats while rice sinks when rice is washed; oil floats on water; sawdust floats but sand sinks. Generally objects lighter than the liquid float and heavier ones sink; the special property that describes this ‘heaviness’ is **density**. Iron feels heavier than a wooden stick of the same size because it is denser. (Density is not the only factor deciding floating or sinking.)

9.5 What Is Density?

FORMULA

Density = Mass ÷ Volume. Density is the mass present in a unit volume of the substance. It does **not depend on shape or size**, but it **depends on temperature and pressure** (pressure matters mainly for gases).

- Analogy: a crowded bus is high density; a dense forest has trees close together.
- **SI unit:** kg/m^3. Other units: g/mL and g/cm^3 (liquids and solids).
- **Conversion:** 1 kg/m^3 = 1000 g/m^3 = 1 g/L = 1 g/1000 mL = 1 g/1000 cm^3 (so 1 g/cm^3 = 1000 kg/m^3).
- 1 mL of water has a mass of about 1 g (at room temperature), so 10 mL water ≈ 10 g and 100 mL ≈ 100 g.
- **Relative density** = density of substance ÷ density of water at that temperature; it has **no units**. Aluminium: 27 g in 10 cm^3 gives 2.7 g/cm^3, i.e. relative density 2.7 (2.7 times as dense as water).
- Oil labelled 1 litre but 910 g shows its density is **less than water**, so oil floats.

▸ 9.5.1 Determining density: measure mass and volume

Quantity	Meaning	Unit / instrument
Mass	Quantity of matter in an object	g, kg; balance
Weight	Force with which Earth attracts an object	newton (N)
Volume	Space occupied	m^3, cm^3 (=cc), dm^3; litre (L) = 1 dm^3; mL = 1 cm^3

NOTE

Most balances (except the **two-pan balance**) actually measure weight but are marked in mass units. Mass is constant; weight changes with place.

Activity 9.3 — Measure mass

Switch on a digital balance, bring it to zero using the **tare** button, place a clean watch glass (or butter paper), tare again, put the stone on it and read its mass (e.g. 16.400 g). For a liquid, use a beaker instead of the watch glass.

Activity 9.4 — Know your measuring cylinder

Maximum volume = largest mark (e.g. 100 mL). **Least count** = volume between two big marks ÷ number of small divisions (10 mL ÷ 10 = **1 mL**). Typical least counts: 0.1 mL for 10 and 25 mL cylinders, 1 mL for 100 mL, 2 mL for 250 mL, 5 mL for 500 mL.

Choosing the cylinder

To measure 70 mL: a 50 mL cylinder needs two steps (inconvenient); 250 or 500 mL cylinders do it in one step but with lower accuracy; a **100 mL cylinder is best**.

Activity 9.5 — Measure 50 mL of water

Place the cylinder on a flat surface, pour slowly to the mark, adjust with a dropper, keep the **eye level with the bottom of the meniscus** (the curved water surface) and read at its bottom. For **coloured liquids read the top** of the meniscus.

Activity 9.6 — Regular solid

Volume of a cuboid = l × w × h (notebook 25 × 18 × 2 = **900 cm^3**).

Activity 9.7 — Irregular solid (displacement)
Take 50 mL water (initial volume A), tie the stone with thread and lower it slowly, note the final level B (55 mL). **Volume of object = B − A = 5 mL = 5 cm^3.** Then density = 16.400 g ÷ 5 cm^3 = **3.28 g/cm^3**.

- Measuring cylinders are **tall and narrow** so that a small change in volume produces a large, easily read change in height (better accuracy than a wide beaker).
- Earth's layers (crust, upper mantle, lower mantle, outer core (liquid), inner core (solid)): **density increases towards the centre** because pressure and temperature rise and material becomes compact. The crust is lightest.
- **Heritage:** bamboo and wooden logs, being light and hollow, were used for rafts and boats in ancient times.

▸ 9.5.2 Effect of temperature on density

On heating, particles move apart, **volume increases but mass stays the same**, so density = mass/volume **decreases**. Cooling raises density. Hence **hot air rises** (less dense than the cool air around it) — the principle of the **hot air balloon**.

▸ 9.5.3 Effect of pressure on density

State	Effect of increasing pressure
Gas	Particles move closer; volume decreases, **density increases** (large effect)
Liquid	Small effect — nearly incompressible
Solid	Even less effect; change usually negligible

IMPORTANT

Why ice floats: water's density is **highest at 4 °C**. When water freezes at 0 °C its particles arrange in a structure taking **more space (expansion)**; same mass, larger volume, lower density, so ice is lighter and floats. The ice layer on top insulates the water below, keeping fish and aquatic animals alive in freezing weather.

Egg test: a raw egg sinks in tap water; dissolving enough **salt** in the water increases the liquid's density until the egg floats. Peeled orange sinks but unpeeled floats because the peel traps air and lowers overall density.

Quick Questions with Model Answers

Likely question	Model answer
Define solute, solvent and solution.	A solution is a uniform mixture; the dissolved substance is the solute and the dissolving medium is the solvent.
In a solution of two liquids, which is the solute?	The one present in the smaller amount.
Distinguish saturated and unsaturated solutions.	Unsaturated can dissolve more solute at that temperature; saturated cannot, and extra solute settles.
What is solubility? How does temperature affect it?	Maximum solute that dissolves in a fixed quantity of solvent at a temperature; for solids it generally increases, for gases it decreases.
Why do fishes die in warm water?	Dissolved oxygen decreases as temperature rises.

Likely question	Model answer
State the formula and SI unit of density.	Density = mass/volume; kg/m3 (g/cm3 also used).
Find the volume of an irregular stone.	Displacement method: final level minus initial level of water in a measuring cylinder.
A stone sculpture of 225 g has volume 90 cm3. Will it float?	Density = 2.5 g/cm3, greater than 1 g/cm3, so it sinks.
Why does hot air rise?	On heating, volume increases while mass is constant so density falls; it is lighter than surrounding cool air.
Why does ice float on water?	Ice has a more spread-out structure, so lower density than liquid water.
What is meniscus and how do you read it?	Curved liquid surface; eye at level, read bottom for colourless liquids and top for coloured liquids.

★ KEY TAKEAWAYS

- A **solution** is a uniform mixture of **solute** and **solvent**.
- **Saturated** solution cannot dissolve more solute at that temperature; **unsaturated** can. Solubility of **solids rises** and of **gases falls** with temperature.
- **Density = mass / volume** (kg/m^3, g/cm^3); relative density has no unit.
- Volume of irregular solids is found by **displacement of water**; read the **meniscus** correctly.
- Heating **decreases** density; pressure greatly **increases** gas density, but barely affects solids and liquids.
- **Ice floats** because water expands on freezing; water is densest at **4 °C**.

CHAPTER 10

Light: Mirrors and Lenses

*A plane mirror always gives an erect image of the same size. Curved surfaces behave differently: they can enlarge, shrink or even invert images, and they can bring light together or spread it out. This chapter covers **spherical mirrors** (concave and convex), the **two laws of reflection**, how these mirrors converge or diverge light, and **lenses** (convex and concave) with their uses.*

Big Questions

- Can we make mirrors that give enlarged or diminished images?
- Why do side-view mirrors say 'Objects in mirror are closer than they appear'?
- Why is there a curved line on some reading glasses?

10.1 What Are Spherical Mirrors?

Activity 10.1 — Shiny spoon
Look into the **inner (curved-in)** side of a spoon: the image is **inverted**. Look at the **outer (bulging)** side: the image is **erect and smaller**. Move the spoon away and see how the image changes.

Meaning
Curved mirrors are not plane mirrors; their image depends on the curve.

DEFINITION

Spherical mirrors have a reflecting surface that is part of a hollow sphere. A **concave mirror** has its reflecting surface curved **inwards**; a **convex mirror** has it curved **outwards**.

- Spherical mirrors are made by grinding and polishing flat glass into a curve, not by slicing a glass sphere.
- Reflective coating (e.g. aluminium) on the **outer** curved surface makes a **concave** mirror; on the **inner** curved surface makes a **convex** mirror.
- In diagrams the non-reflecting (back) side is drawn shaded.
- **Activity 10.2:** place the mirrors on a table with the reflecting surface up and view from the side at eye level — a concave mirror curves inwards (like a bowl), a convex mirror bulges outwards.

10.2 Images Formed by Spherical Mirrors

Activity 10.3: keep a concave and convex mirror upright side by side, place a small toy 3–4 cm away, then slowly move it away, and note size, erect/inverted, and lateral inversion.

Mirror	Object close	Object far away
Plane	Erect, same size	Erect, same size (always)
Concave	**Erect and enlarged**	**Inverted**; first enlarged, then gets smaller as the object moves away
Convex	Erect and diminished	**Always erect and diminished**; becomes slightly smaller as the object moves away

- Image size changes with distance in concave and convex mirrors, unlike plane mirrors.
- **Lateral inversion** (left-right swap) occurs in **all three** types of mirrors.

- So a mirror can be identified as plane, concave or convex just by the image it makes.

▸ Uses of spherical mirrors

Mirror	Use	Reason
Concave	Torch reflectors; headlights of cars and scooters	Concave reflector gives a strong beam
Concave	Dentist's mirror; reflecting telescopes (large main mirror)	Gives an **enlarged** view of teeth; collects light
Convex	Side-view mirrors of vehicles	Erect, diminished image and a **much wider field of view** of the road behind
Convex	Road bends, intersections, shop surveillance	Shows a large area; avoids collisions and deters theft

REMEMBER

The warning 'Objects in mirror are closer than they appear' exists because a convex mirror makes images **smaller**, so vehicles look farther than they really are.

10.3 Laws of Reflection

Activity 10.4 — Reflection from a plane mirror

Use a torch and a comb with all but one slit covered by black paper to get a thin beam along white paper hitting an upright plane mirror. Change the angle of the beam and see the reflected beam shift.

Draw it with rays

Draw the mirror line, the **incident ray** (falls on the mirror) and **reflected ray** (comes back). At the point of incidence O draw a line at **90°** to the mirror — the **normal**.

Measure and compare

The angle between normal and incident ray is the **angle of incidence (i)**; between normal and reflected ray is the **angle of reflection (r)**. Table of readings shows **i = r**. If the beam falls along the normal, i = r = **0°**.

Activity 10.5 — Same plane

Place the mirror on chart paper extending beyond the table edge. The reflected beam is visible on the extended part; **bend the extended part** and the beam disappears, **flatten** and it reappears.

IMPORTANT

Two laws of reflection: (1) the **angle of incidence equals the angle of reflection** (i = r); (2) the **incident ray, the normal at the point of incidence and the reflected ray all lie in the same plane**. These laws hold for **all mirrors — plane, concave and convex**.

▸ Parallel beams and spherical mirrors

Activity 10.6 — Many parallel beams

Leave many comb slits open to get multiple parallel beams and send them on plane, concave and convex mirrors in turn.

Result

Plane: reflected beams stay **parallel**. Concave: reflected beams **converge**. Convex: reflected beams **diverge** — even though each ray still obeys the laws of reflection; the curved surface changes the normal at each point.

Mirror	Effect on parallel beam
Plane	Remains parallel
Concave	**Converges** (brings together)
Convex	**Diverges** (spreads out)

Activity 10.7 — Burning paper with sunlight (teacher present)
Face a concave mirror to the Sun and adjust a sheet of paper until a sharp bright spot forms. The paper smokes and may burn because light is **concentrated** and its heat is large at that point. **Never look at the Sun or its reflection**; focus only on the paper.

Devices that concentrate sunlight with mirrors or lenses are **solar concentrators**; the heat makes steam to generate electricity, for large-scale cooking and in **solar furnaces** that can even melt steel. Solar cookers use mirrors to converge sunlight.

10.4 What Is a Lens?

Activity 10.8 — Water-drop lens
Rub a thin film of oil or wax on a flat glass or plastic strip, place a small water drop on it and put it over printed text. The drop's surface is **curved outwards** and the letters below look **larger** — the drop acts as a simple lens, as does a magnifying glass.

DEFINITION

A **lens** is a transparent piece (glass or plastic) with curved surfaces. A **convex lens** is **thicker at the middle** than the edges; a **concave lens** is **thicker at the edges** than the middle. Light passes **through** a lens, so we see things through it, not in it.

Activity 10.9 — Object behind a lens
Place a small object behind each lens on a stand, look through the lens and slowly increase the distance.

Activity 10.10 — Parallel beams on lenses
Pass many parallel beams through a thin glass plate, convex lens and concave lens: the plate passes them unchanged, the convex lens **converges** them and the concave lens **diverges** them. Hence convex = **converging lens**, concave = **diverging lens**.

Activity 10.11 — Burning paper with a lens
Replace the mirror with a convex lens in sunlight: a bright spot forms and the paper can burn (never look at the Sun through the lens).

Lens	Image of object behind lens	Effect on light
Convex	Close: **erect and enlarged**. Farther: appears **inverted**; first enlarged, then diminished	Converges (converging lens)
Concave	**Always erect and diminished**; size changes with distance	Diverges (diverging lens)
Flat glass	Same size and shape	Passes parallel light unchanged

Feature	Concave mirror	Convex mirror	Convex lens	Concave lens
Light passes through?	No (reflects)	No (reflects)	Yes	Yes

Feature	Concave mirror	Convex mirror	Convex lens	Concave lens
Parallel beam	Converges	Diverges	Converges	Diverges
Image	Erect/enlarged (near), inverted (far)	Always erect, diminished	Erect/enlarged (near), inverted (far)	Always erect, diminished
Uses	Torch, headlights, dentist, telescope	Side-view, road safety, surveillance	Magnifier, spectacles, camera, microscope	Spectacles for some eye defects

Where lenses are used: eyeglasses, cameras, telescopes, microscopes and magnifying glasses. The **human eye has a convex lens** inside that **changes its shape**, letting us focus on near and far objects. A pencil in a half-filled tumbler of water looks changed in shape because the curved, water-filled glass behaves like a lens.

> NOTE
> **Heritage:** over 800 years ago, in the time of Bhāskara II, Indian astronomers used shallow bowls of water and tubes to view reflected images of stars and planets — suggesting practical knowledge of the laws of reflection.

Quick Questions with Model Answers

Likely question	Model answer
What are concave and convex mirrors?	Spherical mirrors whose reflecting surface curves inwards (concave) or outwards (convex).
Describe the image in a convex mirror.	Always erect and diminished, however far the object is.
How does the image in a concave mirror change as an object moves away?	Erect and enlarged when near; becomes inverted farther away, and gets smaller.
State the two laws of reflection.	Angle of incidence = angle of reflection; incident ray, normal and reflected ray lie in one plane.
A ray hits a mirror at 40° to the normal. What angle does the reflected ray make with the mirror?	Angle of reflection 40°, so with the mirror surface it makes 90° − 40° = 50°.
What are the angles of incidence and reflection when light falls along the normal?	Both are 0°.
Why are convex mirrors used as vehicle side-view mirrors?	They give an erect image and a wider field of view.
Why do concave mirrors burn paper in sunlight?	They converge the rays to a point, concentrating heat.
Distinguish convex and concave lens.	Convex is thicker at the middle and converges light; concave is thicker at edges and diverges light.
Why is a magnifying glass a convex lens?	It shows erect, enlarged letters when held close to the text.
Name one use of a concave mirror in a dentist's clinic and why.	Dental mirror gives an enlarged view of teeth.

★ KEY TAKEAWAYS

- **Concave** mirror: curves inwards, converges light, image can be enlarged/diminished and erect/inverted.
- **Convex** mirror: curves outwards, diverges light, image always erect and diminished; wide field of view.
- **Laws of reflection:** i = r; incident ray, normal and reflected ray are in the same plane — valid for all mirrors.
- **Convex lens** converges light (thicker at middle); **concave lens** diverges light (thicker at edges).
- Convex lens image: erect/enlarged when near, inverted when far; concave lens image: always erect and diminished.
- Mirrors and lenses are used in torches, vehicles, spectacles, cameras, telescopes, microscopes, solar cookers and in the eye itself.

CHAPTER 11

Keeping Time with the Skies

*The Moon sometimes shines in the daytime and changes shape from day to day, while the Sun, seasons and Moon give us natural **repeating cycles**. This chapter explains **why the Moon has phases**, where to look for it, how the **day, month and year** led to **lunar, solar and luni-solar calendars**, how festivals follow them, and why we launch **artificial satellites**.*

Big Questions

- Why is the Moon sometimes visible when the Sun is up?
- What would a day, month or year mean on the Moon?
- How could time be measured without clocks or calendars?

11.1 How Does the Moon's Appearance Change, and Why?

Activity 11.1 — Watch the Moon for a month
Start at sunrise on the day after full Moon (Moon in the **west**). Each day record the date, whether seen at sunrise or sunset, shade the bright portion, note if it increased or decreased and whether it is closer to or farther from the Sun. After ~15 days it cannot be seen at sunrise, so observe at **sunset** for the next 15 days.

What you find
The shape differs each day, the Moon is not visible on all days, and its position in the sky relative to the Sun shifts day by day.

11.1.1 Phases of the Moon

- From full Moon the bright part shrinks to a half in about a week and disappears in another week — the **waning period** (**Krishna Paksha**).
- After the new Moon the bright part grows to half in a week and to full in another — the **waxing period** (**Shukla Paksha**).
- **Full Moon day = Purnima** (full bright circle); **new Moon day = Amavasya** (not visible).
- From one full Moon to the next takes about a month. The changing shape of the bright portion seen from Earth is called the **phases of the Moon**.

Phase name	What is seen
New Moon	Not visible
Waxing crescent	Thin bright curve growing
First quarter (half)	Half circle, waxing
Waxing gibbous	More than half bright
Full Moon	Complete bright circle
Waning gibbous	More than half, shrinking
Last quarter (half)	Half circle, waning
Waning crescent	Thin curve shrinking

▸ 11.1.2 Locating the Moon

- On **full Moon**, the Moon is nearly **opposite the Sun**: when the Sun rises in the east, the Moon is nearly setting in the west.
- As the bright part decreases the Moon appears **closer to the Sun** each morning; at **half Moon (waning) it is overhead at sunrise**; the crescent is even closer to the Sun.
- A **waxing Moon is easiest to see at sunset**, and a **waning Moon at sunrise**.
- The Moon rises about **50 minutes later each day**. It does not always rise when the Sun sets; sometimes moonrise is in the afternoon (2–4 p.m.), so the Moon can be seen in the eastern sky in daylight (wait ~30 minutes after the listed time).

▸ 11.1.3 Why do phases occur?

IMPORTANT

The Moon's **shape does not change**; only the part we see changes. The Moon does not emit light but **reflects sunlight**; the half **facing the Sun is illuminated**, the other half is dark. The Moon revolves around Earth in about a month and **always shows the same half to Earth**, but that half is not always the illuminated half. The change in the **fraction of the illuminated portion seen from Earth** causes the phases.

Activity 11.2 — Ball and lamp model

In a dark place, a torch/lamp (~3 m away) represents the Sun, your head the Earth, a ball on a stick the Moon. Hold the ball at arm's length slightly above your head and turn slowly anticlockwise.

Observation

Ball opposite the lamp (A): whole lit part faces you = **full Moon**. Ball toward the lamp (E): only the dark side faces you = **new Moon**. In between, the dividing line is curved, as on the Moon.

Mapping to the orbit

Positions B and H: more than half of the lit part visible = **gibbous**; D and F: less than half = **crescent**; C and G: half visible. A→C→E is the **waning** side, E→G→A is **waxing**.

- Earth's rotation period (one day) is much smaller than the Moon's revolution (~a month), so people everywhere see **nearly the same phase** on a given day.
- New Moon: Moon appears closest to the Sun; full Moon: farthest.
- **Why 50 minutes later?** In one day the Moon moves a bit further in its orbit, so Earth must rotate ~50 minutes more for the Moon to be at the same spot in the sky.

NOTE

Misconception: Earth's shadow does **not** cause phases; Earth's shadow on the Moon causes a **lunar eclipse**. Lunar eclipses occur only on **full Moon**, solar eclipses only on **new Moon**, but not every month because the **Moon's orbit is slightly tilted** relative to Earth's orbit around the Sun.

11.2 How Did Calendars Come into Existence?

Natural periodic events — day and night, Moon phases, seasons — are used to keep time. The Sun's daily rise in the east and set in the west is due to **Earth's rotation** and gives the unit **day**. The average time between the Sun's highest position on two successive days is the **mean solar day = 24 hours**.

Activity 11.3 — Measure a day with a shadow
Fix a 1 m stick vertically on flat ground in sunlight. From ~11:00 a.m. to ~1:10 p.m. mark the tip of the shadow every minute. The **shadow is shortest when the Sun is highest**; note that time. Repeat next day: the difference between times is the solar day (e.g. 12:20 to 12:20 = 24:00; next day 12:19 = 23:59). The average is nearly 24 hours.

Time unit	Natural cycle	Duration
Day	Sun returning to its highest position (Earth's **rotation**)	24 hours
Month	Cycle of Moon's phases (Moon's **revolution** around Earth)	≈ 29.5 days
Year	Cycle of seasons (Earth's **revolution** around the Sun)	≈ 365¼ days

▸ 11.2.1 Lunar calendars

- About **12 lunar months** fit in one cycle of seasons, giving a lunar calendar: shortest unit a day, month ≈ 29.5 days, **lunar year = 12 months = 354 days**.
- Easy to track by watching the Moon, but seasons **do not stay aligned** with the same lunar months each year because the seasons repeat in ~365 days.

▸ 11.2.2 Solar calendars

- Farmers needed to know the seasons, so solar calendars keep the year in step with them. The **Gregorian calendar** is a solar calendar.
- Months are adjusted to add to **365 days** (30- and 31-day months; February 28).
- Earth takes about an **extra quarter day**; these add up to one day every four years, so a **leap year** with a **29-day February** is added when the year is divisible by 4.

REMEMBER
Fine corrections: Earth's year is slightly less than 365¼ days, so century years (1700, 1800, 1900) are **not** leap years, but a year divisible by **400** (1600, 2000) is a leap year again. The Gregorian calendar follows the **tropical year** (successive spring equinoxes); the **sidereal year** (same stars rising at sunset) is longer by about **20 minutes** and is used by astronomers to track Earth's position in its orbit.

Heritage: ancient observers, without knowing Earth goes round the Sun, found the year ≈ 365 days. The Sun rises a little north of east in summer and a little south of east in winter, the extremes being the **solstices** (~21 June and 21 December). The Sun's apparent northward movement (December to June) is **Uttarayan**, southward (June to December) is **Dakshinayan**. The Taittirīya Saṁhitā (6.5.3) states the Sun moves south for six months and north for six months, and the Surya Siddhanta ties these to the Sun's entry into Makar (Capricorn) and Karka (Cancer). On equinoxes the Sun rises exactly east and sets exactly west.

▸ 11.2.3 Luni-solar calendars

- They count days and months by the Moon's phases but adjust to stay in step with seasons.
- 12 lunar months = 354 days, **about 11 days short** of the solar year; in 2–3 years the gap is nearly a month, so an extra month, **Adhika Maasa (intercalary month)**, is added every few years.
- Months: Chaitra, Vaisakha, Jyeshtha, Ashadha, Shravana, Bhadrapada, Ashwin, Kartika, Margashirsha, Pausha, Magha, Phalguna.
- **Amant** calendars: month starts after new Moon, ends on new Moon. **Purnimant**: month starts after full Moon, ends on full Moon.

▸ 11.2.4 The Indian National Calendar

- A **solar** calendar of **365 days**, used with the Gregorian calendar for official purposes; year starts on **22 March** (day after spring equinox).
- Months have 30 or 31 days: in a regular year months 2–6 have 31 days and the rest 30; names from traditional Indian calendars (first month **Chaitra**).
- In leap years a day is added to Chaitra and the year begins on 21 March.
- The **Calendar Reform Committee** (1952) recommended it; adopted **21 March 1956** (1 Chaitra 1878 Saka); follows Surya Siddhanta principles.
- **Meghnad Saha** (1893–1956), astrophysicist known for the Saha equation, chaired the committee; the Saha Institute of Nuclear Physics, Kolkata, is named after him.

Calendar	Based on	Features	Example
Lunar	Moon phases	354-day year; seasons drift	Islamic calendar (Eid-ul-Fitr)
Solar	Earth's revolution / seasons	365 days + leap year	Gregorian, Indian National Calendar
Luni-solar	Moon phases + seasons	Intercalary (Adhika) month	Many Indian regional calendars

11.3 Are Festivals Related to Astronomical Phenomena?

- Many festivals follow Moon phases: **Diwali** on the new Moon of Kartika, **Holi** on the full Moon of Phalguna, **Buddha Purnima** on the full Moon of Vaisakha, **Dussehra** on the 10th day of Ashwina; **Eid-ul-Fitr** is celebrated after sighting the **crescent Moon** at the end of Ramazan.
- They fall on different Gregorian dates each year. Luni-solar festivals shift by **less than a month** because of the intercalary month; purely lunar festivals like Eid move through all months (about **11 days earlier** each year).
- **Makar Sankranti, Pongal, Bihu, Vaisakhi, Poila Baisakh, Puthandu** follow a **solar sidereal** calendar and fall on almost the same Gregorian date; slow wobble of Earth's axis shifts them gradually (Makar Sankranti moves one day every 71 years).
- Festival dates depend on lunar phase at sunrise, which differs by a day between east and west India; the **Positional Astronomy Centre** publishes the **Rashtriya Panchang** for uniform dates.
- **Culture:** Moon imagery appears in ragas (Chandrakauns, Chandranandan), Bharatanatyam mudras (Chandrakala, Ardhachandran), Madhubani and Warli paintings, Dhokra art.

11.4 Why Do We Launch Artificial Satellites?

- The Moon is Earth's **natural satellite**; **artificial satellites** are human-made and orbit Earth, appearing as fast-moving specks. Most orbit ~**800 km** above Earth and take about **100 minutes** per orbit — also a periodic motion.
- Uses: **communication, navigation, weather monitoring, disaster management, scientific research**.
- ISRO: **Cartosat** (images for maps and city planning; **Bhuvan** platform), **AstroSat** (observes stars), Chandrayaan 1, 2, 3 (Moon), Aditya L1 (Sun), Mangalyaan (Mars); student satellites AzaadiSat, InspireSat-1, Jugnu.
- **Activity 11.4:** spot satellites just before sunrise or after sunset (with an adult, clear sky): a steady or flickering point of light moving fast; apps help predict passes.
- **Space debris:** dead satellites and rocket parts crowd space and may collide with working satellites; small pieces burn in the atmosphere, large ones can reach the ground.

- **Vikram Sarabhai** (1919–1971), Father of the Indian Space Programme; Vikram Sarabhai Space Centre (Thiruvananthapuram) develops rockets and launch vehicles.

> **NOTE**
>
> **Tides:** sea level rises and falls in a regular pattern; a given tide returns about **50 minutes later each day**, like moonrise, showing tides are linked to the Moon's position and phase. The crescent's bright side always faces the Sun.

Quick Questions with Model Answers

Likely question	Model answer
What causes the phases of the Moon?	As the Moon revolves around Earth, we see different fractions of its sunlit half.
Do Earth's shadows cause Moon phases?	No; Earth's shadow causes a lunar eclipse.
Differentiate waxing and waning, Shukla and Krishna Paksha.	Waxing: bright part grows (Shukla Paksha); waning: it shrinks (Krishna Paksha).
When is a waxing Moon best seen? A waning Moon?	Waxing at sunset; waning at sunrise.
Why does the Moon rise ~50 minutes later each day?	It moves ahead in its orbit, so Earth must rotate further.
Why don't eclipses occur every month?	The Moon's orbit is slightly tilted to Earth's orbit.
On what natural cycle are day, month and year based?	Earth's rotation, Moon's phases (revolution), and the cycle of seasons (Earth's revolution).
Why is a leap year needed?	Earth takes about an extra quarter day; four years add one day.
Why do lunar-calendar festivals fall on different dates?	A lunar year has only 354 days; luni-solar calendars add an intercalary month.
What is the Indian National Calendar?	A 365-day solar calendar starting 22 March, adopted 21 March 1956.
Give uses of artificial satellites.	Communication, navigation, weather monitoring, disaster management and research.
Amol was born on a full Moon day on 6 May. Is his birthday on full Moon every year?	No; the Moon cycle (29.5 days) does not match the Gregorian calendar.

★ KEY TAKEAWAYS

- The Moon shines by **reflected sunlight**; its shape is constant, but the lit fraction we see changes — **phases** (~29.5 days).
- Full Moon is opposite the Sun; new Moon is invisible; **waxing** = Shukla Paksha, **waning** = Krishna Paksha.
- Moonrise is ~**50 minutes later daily**; Earth's shadow gives eclipses, not phases.
- **Day** (rotation), **month** (Moon's phases), **year** (seasons) are the basic time units.
- **Lunar** (354 days), **solar** (365 days + leap year) and **luni-solar** (with Adhika Maasa) calendars; Indian National Calendar starts 22 March.
- Festivals follow lunar, luni-solar or solar-sidereal calendars; **artificial satellites** serve communication, navigation, weather and research.

CHAPTER 12

How Nature Works in Harmony

*When forests shrink and waterholes dry, elephants wander into farms — one change leads to another. This chapter shows how living and non-living parts of a **habitat** form an **ecosystem**, how organisms are linked through **food chains and food webs**, how **decomposers** recycle nutrients, how **interactions keep nature balanced**, and how human activity, especially farming, can disturb that balance.*

Big Questions

- How might loss of forest cover and changed rainfall lead elephants into farms and villages?
- Can the Earth thrive without humans? Can humans survive without the Earth?
- Can human actions cause natural disasters?

Elephant example: in Odisha, Jharkhand, West Bengal, Assam and Chhattisgarh, elephants enter farms and villages when vegetation is scarce and waterholes dry up, seeking bananas and sugarcane. Climate change in rainfall/temperature and felling of trees for roads and buildings shrink forests; with **habitat loss** animals move into human areas, causing crop damage and harm. **Elephant corridors** connect forest habitats so animals can move safely without conflict.

12.1 How Do We Experience and Interpret Our Surroundings?

DEFINITION

A **habitat** is the place where an organism lives (even tree bark). It has **biotic components** (living: plants, animals, microorganisms) and **abiotic components** (non-living: air, sunlight, water, temperature, soil).

Habitat	Biotic components	Abiotic components
Pond	Fish, frogs, turtles, snakes, dragonflies, mosquitoes, snails, ducks, algae, diatoms, duckweed, lotus	Water, sunlight, oxygen in water, soil, temperature
Forest	Grass, trees, birds, deer, microbes	Soil, air, sunlight, water, temperature

- **Activity 12.1:** list living and non-living things of two habitats (pond, forest, farm, large tree) in groups with a teacher. Both have living and non-living parts; the types differ.
- A pond gives fish food, oxygen, shelter and space: food from small plants and animals (biotic), oxygen from water (abiotic).
- Different organisms share resources differently — a nocturnal snake and a day-active rodent live in the same forest under different conditions, so they **coexist**.

12.2 Who All Live Together in Nature?

Term	Meaning	Example
Population	Group of the **same kind** of organisms living in a habitat at a given time	All the fish of one kind in a pond; 20 plants of one kind in a 1 m × 1 m patch

Term	Meaning	Example
Community	All the **different populations** (plants, animals, microorganisms) sharing a habitat	Fish, frogs, algae, insects of a pond
Ecosystem	Community + abiotic components interacting	Pond, forest, farm

- **Activity 12.2:** mark a 1 m × 1 m area of the school garden, identify four organisms, count them (population) and pool group data.
- If all organisms were the same they would need the same food, water and space, causing **competition and scarcity**.
- **Pollination:** flower has sepals, petals, **carpels (female)** and **stamens (male, release pollen)**. Wind, water, insects, bats and birds carry pollen — essential for fruits and seeds.

12.3 Does Every Organism in a Community Matter?

Activity 12.3 — Pond study: Pond A (with fish) had many flowering plants nearby; Pond B (no fish) had fewer. Fish eat **dragonfly larvae**, so Pond A had fewer dragonflies; dragonflies eat flies, bees and butterflies, so Pond A had **more pollinators**, hence more pollination and **more seeds** in nearby flowers.

> **IMPORTANT**
> This shows **direct and indirect effects**: every organism matters, and biotic (fish, dragonflies, pollinators, plants) and abiotic (temperature, water, nutrients) parts affect one another. Overfishing could upset this balance.

12.4 Interactions Among Organisms and Their Surroundings

Criterion	Kind of interaction	Example
1	Between **biotic and abiotic** components	Earthworms live in moist soil; a fish lays eggs in water; microbes in pond water
2	Between **two abiotic** components	Sunlight raises temperature and evaporates water fast; air currents make waves
3	Among **biotic** components	Frog eats insects; water snake eats fish; frog and fish compete for larvae

Biotic and abiotic components interacting form an **ecosystem**. Ecosystems can be large or small and may overlap:

Type	Examples
Aquatic	Ponds, rivers, lakes, seas, oceans
Terrestrial	Forests, grasslands, deserts, farms, large trees (banyan, mango, pilkhan)
Human-made	Farmland, fish ponds, parks

- **Biotic depend on abiotic:** sunlight, CO_2 and water for making food; soil gives nutrients and support; air gives oxygen for respiration; water is essential for all.
- **Abiotic are influenced by biotic:** plants release oxygen, roots hold soil and prevent erosion, plants retain soil moisture and help cool the atmosphere.

▸ Feeding habits

Group	Meaning	Examples
Producers (autotrophs)	Make own food by photosynthesis (auto = self, troph = food)	Green plants, trees, grass
Consumers (heterotrophs)	Depend on other organisms for food (hetero = other)	Animals
Herbivores	Eat only plants	Deer, hare
Carnivores	Eat only animals	Leopard, tiger, vulture
Omnivores	Eat plants and animals	Crow, fox, mouse
Decomposers (saprotrophs)	Break down dead matter (sapro = rotten)	Fungi (mushroom), bacteria

12.5 Who Eats Whom?

- A **food chain** is a simple linear sequence showing 'who eats whom': *Grass* → *Grasshopper* → *Frog* → *Snake* → *Eagle*; the arrow points to the eater.
- Each step is a **trophic level**: producers (1st), herbivores (2nd), small carnivores such as frogs (3rd), large carnivores such as tigers or vultures (next).
- **Pyramid (Activity 12.7):** numbers are highest at the base (millet) and lowest at the top (eagle), giving a pyramid shape.
- Organisms are eaten by two or more kinds of animals, so chains interlink into a network — a **food web** (Activity 12.8: hawk, snake, fox, owl, hare, mouse, frog, bird, grasshopper, grasses).

12.6 What Happens to Waste in Nature?

- Dead plants/animals and waste are broken down by fungi and bacteria (mushrooms grow on dead matter in rains; beetles and flies on dung) — **decomposition**.
- Complex substances become simple ones and **nutrients return to the soil** for plants. Decomposers (saprotrophs) therefore **recycle nutrients** — in nature nothing is wasted.
- **Migratory birds** (e.g. Demoiselle cranes at Khichan, Jodhpur, in winter) move to avoid harsh climate and find food, act as pollinators and seed dispersers, link habitats and eat insect pests, helping farmers.

12.7 How Does One Change Lead to Another?

Chain reaction in a pond
Pollution kills pond plants → **less oxygen** → fish die → fewer consumers → insects increase → spread to farms → farmers use more **pesticides** → further environmental damage.

Activity 12.9 — Indian bullfrog
In the 1980s India exported large numbers of **frog legs**; frog populations fell, frogs' insect prey (agricultural pests) rose, farmers used more synthetic pesticides that harmed soil, water and health. The government **banned frog-leg export**.

REMEMBER
An ecosystem is in **balance** when interactions keep populations and resources stable. This balance is **dynamic, not fixed** and can be upset by natural or human-made changes.

12.8 How Do Interactions Maintain Balance?

Organisms **compete** for food, water, space or sunlight; competition limits population size and prevents one species from multiplying too much. Other relationships:

Relationship	Effect	Example
Mutualism	Both benefit	Honeybee gets nectar, flower gets pollinated
Commensalism	One benefits, the other unaffected	Orchid on a tree gets support; the tree branch is unaffected
Parasitism	One benefits, the other harmed	Tick feeds on dog's blood; dog gets skin irritation

> **NOTE**
> **A.J.T. Johnsingh**, wildlife biologist, pioneered modern tracking and showed (in Bandipur) that predators like tigers and leopards depend on a healthy prey population such as deer and wild boar.

12.9 Benefits of an Ecosystem

- Forests give fresh air, fertile soil, food, fibre, timber and medicines; aquatic ecosystems give water and food; ecosystems also have aesthetic and recreational value and regulate climate.
- **Sundarbans:** world's largest mangrove forest where Ganga and Brahmaputra meet (India–Bangladesh); slows winds and waves in storms and floods, absorbs CO_2 and gives oxygen; **UNESCO World Heritage Site (1987)**.
- **Threats:** cutting mangroves for fuelwood and farming, illegal hunting, overuse, industrial waste and sewage. Mangroves also protected Selvam's village in the 2004 tsunami by reducing wave force.
- General threats: deforestation, overuse of resources, **invasive species**, unsustainable land use, pollution, climate change.
- **Protected areas:** national parks, wildlife sanctuaries, biosphere reserves, community conserved areas — e.g. Jim Corbett (Uttarakhand), Manas (Assam), Nilgiri Biosphere Reserve, Chilika Lake (Odisha), Eaglenest (Arunachal Pradesh), Hemis (Leh), Keibul Lamjao (Manipur), Pirotan Island Marine (Gujarat).

▸ 12.9.1 Human-made ecosystems

Fish ponds, farms and parks meet human needs; if well designed they reduce pollution, support biodiversity and give recreation, but unlike natural ecosystems they **need human care and management**.

▸ 12.9.2 Healthy ecosystems and farming

- Between 1950 and 1965 India faced a food crisis; tractors, machines, **synthetic fertilisers and pesticides** raised production — the **Green Revolution**.
- These methods are now considered unsustainable: **overuse of chemicals, excessive groundwater extraction, monoculture** (growing one crop repeatedly).
- Overusing fertilisers kills useful soil microorganisms and reduces **humus**, so soil erodes; pesticides kill natural predators (e.g. beetles that eat pests) so pests increase; pests become **resistant**; heavy irrigation and ploughing disturb earthworms and snails; monoculture reduces biodiversity and pollinators.
- Solution: **organic and natural farming**, minimal interference, crop variety, recycling farm waste.
- **Heritage:** the *Vrikshayurveda* advocates soil nourishment with organic manure such as **Kunapa Jala**, a fermented liquid fertiliser from animal and plant waste.

Quick Questions with Model Answers

Likely question	Model answer
Define habitat, biotic and abiotic components.	Habitat: place where an organism lives. Biotic: living components; abiotic: non-living such as air, water, soil, sunlight.
Distinguish population, community and ecosystem.	Population: same kind of organisms; community: all populations in a habitat; ecosystem: community plus abiotic factors interacting.
Name the two types of ecosystems with examples.	Aquatic (pond, river, lake, sea) and terrestrial (forest, grassland, farm).
Who are producers, consumers and decomposers?	Producers make food by photosynthesis; consumers eat plants/animals; decomposers (bacteria, fungi) break down dead matter.
What is a food chain and a food web?	A food chain is a linear 'who eats whom' sequence; food webs are interlinked chains.
What are trophic levels?	Positions of organisms in a food chain.
Why are decomposers essential?	They recycle nutrients to the soil; otherwise dead matter piles up and plants lack nutrients.
Distinguish mutualism, commensalism and parasitism with examples.	Both benefit (bee–flower); one benefits other unaffected (orchid on tree); one benefits other harmed (tick on dog).
How did frog-leg export affect farming?	Fewer frogs meant more insect pests, more pesticide use and environmental harm.
Why are mangroves important?	They slow winds and waves in storms, absorb CO2 and shelter wildlife.
Why is monoculture harmful?	It reduces biodiversity and pollinators and degrades soil.
What happens to grasshoppers and snakes if frogs disappear?	Grasshoppers increase (no predator) and snakes decrease (less food).

★ KEY TAKEAWAYS

- A **habitat** has **biotic** and **abiotic** components; their interaction forms an **ecosystem** (aquatic or terrestrial).
- **Population → community → ecosystem** go from smaller to larger.
- **Producers** make food, **consumers** eat plants/animals, **decomposers** recycle nutrients.
- **Food chains** link into **food webs**; positions are **trophic levels**.
- **Mutualism, commensalism, parasitism** and **competition** keep balance; small changes can cascade.
- Human activities (deforestation, pollution, monoculture, overuse of chemicals) threaten ecosystems; **protected areas** and sustainable farming conserve them.

CHAPTER 13

Our Home Earth, a Unique Life-Sustaining Planet

*Among perhaps billions of planets, Earth is the only one known to support life in all its forms. This closing chapter explains **what makes Earth suitable for life** (distance, size, atmosphere, ozone, magnetic field), how **air, water, soil and living things** work together, how **reproduction** keeps life going while allowing change, and the **threats** — the triple planetary crisis — that human activity creates.*

Big Questions

- What would Earth look like with no life at all?
- What lets life survive for billions of years despite disasters?
- If a spaceship carried soil and water to Mars, could plants start growing?

13.1 Why Is Earth a Unique Planet?

- All mountains, rivers, forests, animals and people exist in a **very thin layer** at the surface. If Earth were an apple, the crust would be as thin as the apple's skin.
- ISRO satellite images (this one was a mosaic of ~3000 images, in **false colour**) help study land plants, ocean organisms, ocean temperature, oil spills and wind direction.
- Earth gives air to breathe, water to drink, soil for crops and rock, timber for building. Air does not escape into space (gas particles are held by gravity); gravity holds us on the ground though our heart still pumps blood to the head.

13.2 The Planets of the Solar System

- Eight planets move around the Sun in nearly circular orbits: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune** (in order of increasing distance).
- **Mercury, Venus, Earth, Mars** are small and **rocky**; **Jupiter, Saturn, Uranus, Neptune** are large and made mostly of **gases**.
- Activity 13.2 compares radius (relative to Earth), average temperature and atmosphere: Mercury ≈ 170 °C, no atmosphere; Venus ≈ 450 °C, radius 0.95 of Earth, atmosphere; Earth ≈ 15 °C, atmosphere; Neptune ≈ −200 °C.

IMPORTANT

Venus is the hottest planet, though Mercury is closer to the Sun, because its thick atmosphere (almost entirely **carbon dioxide**) traps heat — the **greenhouse effect**. On Earth too, gases such as CO_2 absorb the radiation given off by the Sun-warmed Earth and keep it at the right temperature.

	Greenhouse effect (planets)	Greenhouse for plants
How it works	Atmospheric gases **absorb radiation** from the warmed surface	Closed space with glass walls **traps warm air**
Example	Venus, Earth	Plant house in a cool climate

13.3 What Makes Earth Suitable for Life?

▸ 13.3.1 Position of the Earth — the habitable zone

- Earth is at the **right distance from the Sun**, so water stays **liquid**. Closer: too hot, water evaporates. Farther: too cold, water freezes. From what we know, liquid water is essential for life to evolve (though some microbes survive in ice).
- The range of distances from a star where water stays liquid is the **habitable zone** or **Goldilocks zone**.
- Water covers most of Earth's surface, so Earth looks blue from space — the **Blue Planet**.
- **Mars** lies at the edge of the habitable zone; no proof of life yet but it may once have had liquid water and lakes. Science stays open to new evidence.

▸ 13.3.2 Size of the Earth

- Earth's orbit is **almost circular**, so sunlight and heat stay nearly steady through the year (no extreme summers and winters at most places).
- Earth is the right size for gravity to **hold an atmosphere**. A much smaller Earth (same density) would have weak gravity and lose its gases to space; **Mars' atmosphere is ~100 times thinner** and **Mercury has none**. A much bigger planet would have gravity strong enough to crush bones.
- **Oxygen** is needed for breathing; some becomes **ozone** (three-atom oxygen molecule, O_3), forming the **ozone layer** that blocks harmful **ultraviolet (UV)** rays.
- **Mangalyaan** (Mars Orbiter Mission, ISRO, 2013) studied Mars' atmosphere, surface and signs of past water with low-cost technology.

▸ 13.3.3 Magnetic field of the Earth

- Earth behaves like a giant magnet (a suspended magnet settles in a fixed direction); the field probably arises from **moving molten iron in the core**.
- Earth is hit by high-energy particles: **cosmic rays** (from far space) and **solar wind** (from the Sun), which could damage the atmosphere, reduce ozone and let in more UV.
- The magnetic field acts as a **protective shield**, pushing many of these particles away.

Factor	Role in making Earth habitable
Distance from Sun	Liquid water; moderate temperature
Nearly circular orbit	Steady sunlight through the year
Size / gravity	Holds the atmosphere; not crushing
Atmosphere (oxygen, greenhouse gases)	Breathing, warmth, protection
Ozone layer	Blocks harmful UV rays
Magnetic field	Deflects cosmic rays and solar wind

13.4 What Allows Life to Be Sustained?

▸ 13.4.1 Air, water and sunlight

- Oxygen from air is used in respiration; in sunlight, plants use CO_2 and water for **photosynthesis**, releasing oxygen.
- The mild **greenhouse effect** keeps temperature high enough for liquid water; without an atmosphere Earth would lose heat to space and be too cold.

- Water covers about **70%** of the surface and forms the **hydrosphere** (ponds, lakes, rivers, springs, seas, oceans, groundwater). It is a good solvent, carries nutrients in plants and regulates temperature, aids digestion and hydration in animals; its vapour forms clouds and rain, refilling rivers, lakes and groundwater. Moving air shapes weather and rainfall.

▸ 13.4.2 Soil, rocks and minerals

- The solid Earth — rocks, soil, minerals — is the **geosphere**; it gives salt, coal, oil and metals like iron and copper.
- Soil is rich in nutrients such as **nitrogen and potassium** from slowly weathered rocks and remains of plants and animals.
- The variety of landforms, rocks and soils and the processes that shape them is **geodiversity**, which creates unique habitats.

▸ 13.4.3 Plants, animals and microorganisms — the biosphere

All living beings together with the places they live make up the **biosphere** (land, water, air). Plants make food, animals eat plants/animals, decomposers recycle nutrients.

Sphere	Meaning
Atmosphere	Air (layer of gases around Earth)
Hydrosphere	All water
Geosphere	Solid Earth: rocks, soil, minerals
Biosphere	All life and its habitats

▸ 13.4.4 The importance of balance

Land, air, water and living things form one connected system. A small change — such as cutting a forest — can affect rainfall, soil, air quality and animals. Life survives because **everything works together in balance**, so protecting clean air, water, soil and life is essential.

13.5 What Keeps Life from Disappearing? — Reproduction

- Reproduction ensures each type of organism continues. Parents pass **genetic material (genes)** — an instruction manual inside every cell — so a calf grows into a cow and a kitten into a cat.
- Reproduction also allows **small changes** (variations), which sometimes help survival: camels' humps store fat for deserts; bacteria evolve resistance to antibiotics. Over generations changes can produce new features or new types of organisms.

	Asexual reproduction	Sexual reproduction
Parents	**One**	**Two** (male and female; even bacteria and yeast have two 'mating types')
Offspring	**Exact copies** of the parent	Mixture of both parents' genes — similar but **not identical** (variation)
Examples	Vegetative propagation; binary fission in bacteria and amoeba; budding in hydra; regeneration in planaria	Flowering plants, fish, frogs, birds, mammals, humans

▸ 13.5.1 Asexual reproduction

- **Vegetative propagation:** a part of a plant — leaf, stem or root — planted in soil grows into a new plant (money plant stem cutting, ‘eyes’ of potato, ginger, bamboo, sugarcane).
- **Activity 13.3:** plant a money plant cutting, potato eyes or ginger piece in moist soil with water, air and sunlight; note days taken for roots, stem and the first leaf.
- Single-celled **bacteria and amoeba divide into two identical individuals**; algae regrow from cut parts; **hydra** grows **buds** that break off and become new individuals; **planaria** (flatworm) can regrow from a small fragment — studied to understand **regeneration**.

▸ 13.5.2 Sexual reproduction

- Each parent makes special reproductive cells, **gametes**, carrying **half** the genetic material; when male and female gametes join, a new cell with a complete set forms (so the amount of instructions does not keep doubling).
- Each gamete carries a different mix of instructions (eye colour, hair type…), so every child is unique — even brothers and sisters differ.

	Plants	Animals
Male gamete	**Pollen grains** (in the anther)	**Sperm**
Female gamete	**Ovule** (deep inside the flower)	**Egg**
Transfer	**Pollination** by wind, insects or animals	In water, sperm and eggs are released (fish, frogs); in birds and mammals sperm are deposited inside the female
Joining	**Fertilisation** gives a **zygote** that becomes the **seed**; the fleshy part around the ovule becomes the **fruit**	Fertilisation gives a **zygote** which develops into an **embryo**
Spread/Development	Seeds spread by birds/animals eating fruits (banyan seed in a wall crack); with water they use stored food to grow roots and shoots	Fish/frogs: development in water. Birds: egg laid, embryo develops outside with food stored in egg. Most mammals: embryo develops **inside the mother**, who supplies food and oxygen

13.6 Threats to Life on Earth

> **IMPORTANT**
> Climate change, biodiversity loss and pollution are together called the **triple planetary crisis**.

- **Climate change:** burning coal and oil releases extra CO_2 and methane (greenhouse gases) that were locked underground for millions of years; Earth cannot absorb them fast enough, so **global warming** occurs. Result: melting ice caps, rising sea levels (flooding coastal cities), extreme weather, loss of species. Long-term changes in temperature, rainfall and weather patterns = **climate change**.
- **Biodiversity loss:** habitat destruction removes species; e.g. if grasses vanish, deer and grasshoppers suffer and then predators like tigers and foxes. Every organism has a role.
- **Pollution:** air pollution from factories, vehicles and burning fuels causes breathing problems, crop damage, **smog and acid rain**; water pollution (factory, farm and plastic waste) harms aquatic life; excess fertilisers and poor waste disposal pollute soil, cut yields and spread harmful substances through the food chain.
- A small change in oxygen, temperature or ozone could endanger life.

Global agreement	Aim / outcome
Montreal Protocol (1987)	Reduced CFCs; ozone layer slowly recovering
Earth Summit (1992)	International efforts on climate change and biodiversity
Kyoto Protocol (2005)	Countries committed to cut greenhouse gases
Paris Agreement (2015)	Limit warming to below 1.5 °C (world not on track as of 2025)

▸ What can we do?

- Cut greenhouse gases using **renewable energy** (solar, wind), save energy, choose green travel.
- Preserve biodiversity; local communities use resources sustainably.
- Better waste management and sustainable farming to protect water and soil.
- **Reuse, repair, recycle**; save energy and water; learn and share ideas.

Quick Questions with Model Answers

Likely question	Model answer
What is the habitable zone?	The range of distances from a star where water can stay liquid; also called Goldilocks zone.
Why is Venus hotter than Mercury?	Its thick CO2 atmosphere traps heat (greenhouse effect).
Why does Earth hold an atmosphere?	Its size gives enough gravity; a smaller planet would lose its gases to space.
What is the ozone layer and why is it important?	A layer of ozone (O3) that blocks harmful UV rays from the Sun.
How does Earth's magnetic field protect life?	It deflects cosmic rays and solar wind that could damage the atmosphere and ozone.
Name the four spheres of the Earth system.	Atmosphere, hydrosphere, geosphere and biosphere.
What is geodiversity?	The variety of landforms, rocks and soils and the processes that shape them.
Differentiate asexual and sexual reproduction.	Asexual: one parent, identical copies; sexual: two parents, offspring with mixed traits.
What are gametes? Why do they carry half the genetic material?	Special reproductive cells; so the zygote gets a full set and the amount does not double each generation.
Define pollination and fertilisation.	Pollination: transfer of pollen to another flower; fertilisation: joining of male and female gametes forming a zygote.
What is the triple planetary crisis?	Climate change, biodiversity loss and pollution.
List five examples of vegetative propagation.	Money plant stem cutting, potato eyes, ginger, bamboo, sugarcane.

★ KEY TAKEAWAYS

- Earth is unique: **right distance** (liquid water), **nearly circular orbit**, **right size** (atmosphere), **ozone layer**, **magnetic field**.
- Venus is hottest due to the **greenhouse effect** of its thick CO_2 atmosphere.
- **Atmosphere, hydrosphere, geosphere and biosphere** interact to sustain life; balance is vital.
- **Reproduction** continues life; **asexual** gives exact copies, **sexual** (gametes → zygote) gives variation.
- **Climate change, biodiversity loss and pollution** form the **triple planetary crisis**; renewable energy, conservation and 3 Rs help.